* Solutions *

1. (12 points) Find the equation of the tangent plane to $f(x,y) = \sin(xy)$ at the point $\left(\frac{2\pi}{3}, \frac{1}{2}\right)$.

   Answer: $z = \frac{\sqrt{3}}{2} + \frac{1}{4}\left(x - \frac{2\pi}{3}\right) + \frac{\pi}{3}\left(y - \frac{1}{2}\right)$

   Show work here:

   $f\left(\frac{2\pi}{3}, \frac{1}{2}\right) = \sin\left(\frac{\cancel{2}\pi}{3} \cdot \frac{1}{\cancel{2}}\right) = \sin(\pi/3) = \frac{\sqrt{3}}{2}$

   $f_x = y\cos(xy) \Rightarrow f_x\left(\frac{2\pi}{3}, \frac{1}{2}\right) = \frac{1}{2}\cos(\pi/3) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

   $f_y = x\cos(xy) \Rightarrow f_y\left(\frac{2\pi}{3}, \frac{1}{2}\right) = \frac{2\pi}{3}\cos(\pi/3) = \frac{2\pi}{3} \cdot \frac{1}{2} = \frac{\pi}{3}$

2. (9 points) Estimate $\Delta f = f(1.05, 0.96) - f(1,1)$ assuming that $f(1,1) = 1$, $f_x(1,1) = 3$, $f_y(1,1) = -4$. Simplify your answer.

   Answer: $\Delta f \approx 0.31$

   Show work here:

$$f(1.05, 0.96) \approx L(1.05, 0.96) = 1 + 3(1.05 - 1) - 4(0.96 - 1)$$
$$= 1 + 3(0.05) - 4(-0.04)$$
$$= 1 + 0.15 + 0.16$$
$$= 1.31$$

$$\Delta f = f(1.05, 0.96) - f(1,1) \approx L(1.05, 0.96) - f(1,1) = 1.31 - 1 = 0.31$$

or use

$$\Delta f = f(x,y) - f(a,b) \approx f_x(a,b)\Delta x + f_y(a,b)\Delta y$$

3. (12 points) Let $f(x,y,z) = y^2 - 2xy - z^2$, $\mathbf{v} = \langle 3,0,4\rangle$, and $P = (2,1,-1)$.

(3.1) a. A unit vector that points in the direction of the maximum rate of increase f at P is $\langle -1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}\rangle$ or $\langle -2/\sqrt{12}, -2/\sqrt{12}, 2/\sqrt{12}\rangle$.

b. The maximum rate of increase of f at P is $2\sqrt{3}$ or $\sqrt{12}$.

Show all work:

$\nabla f = \langle -2y, 2y-2x, -2z\rangle$

$\nabla f_P = \nabla f(2,1,-1) = \langle -2,-2,2\rangle$

$\|\nabla f_P\| = \sqrt{(-2)^2+(-2)^2+2^2} = \sqrt{12} = 2\sqrt{3}$

(3.2) Find the directional derivative of f in the direction of $\mathbf{v}$ at the point P . Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

(A) 5

(B) $\frac{2}{25}$

(C) $\langle -2,-2,2\rangle$

(D) $\langle 3,0,4\rangle$

(E) $\left\langle -\frac{6}{5}, 0, \frac{8}{5}\right\rangle$

(F) $\left\langle -\frac{6}{25}, 0, \frac{8}{25}\right\rangle$

(G) $2\sqrt{3}$

(H) $\frac{2}{5}$ (filled)

(I) $\langle 0,0,0\rangle$

(J) $\langle 2,1,-1\rangle$

(K) 2

(L) $\frac{1}{\sqrt{3}}$

$\vec{u} = \dfrac{\vec{v}}{\|\vec{v}\|} = \dfrac{\langle 3,0,4\rangle}{\sqrt{9+0+16}} = \langle 3/5, 0, 4/5\rangle$

$D_{\vec{u}} f(P) = \langle -2,-2,2\rangle \cdot \langle 3/5, 0, 4/5\rangle$

$= -\frac{6}{5} + 0 + \frac{8}{5}$

$= 2/5$

4. (9 points) Let $f(x,y)$ be such that $\nabla f(x,y) = \langle 3x^2y, x^3\rangle$, and let $x(s,t) = s^2$ and $y(s,t) = st^2$. Apply the multivariable calculus chain rule to calculate $\frac{\partial f}{\partial s}$. Write only your final answer in the box provided. Your final answer must be simplified and written in terms of s and t.

$$\frac{\partial f}{\partial s} = \boxed{7s^6t^2}$$

Show your work here:

$$\frac{\partial x}{\partial s} = 2s, \quad \frac{\partial y}{\partial s} = t^2$$

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}$$

$$= 3x^2y(2s) + x^3(t^2)$$

$$= 3(s^2)^2(st^2)(2s) + (s^2)^3t^2$$

$$= 6s^6t^2 + s^6t^2$$

$$= 7s^6t^2$$

5. (6 points) Use implicit differentiation to find $\frac{\partial z}{\partial x}$ for $z^3 + zx + x^2y = x + 2yz$. Fill in **one** bubble that corresponds to your answer. Questions with multiple answers selected will receive no credit. No work required.

(A) $-\frac{z+2xy}{3z^2+x}$

(B) $-\frac{3z^2+x-2y}{z+2xy-1}$

(C) $\frac{x^2-2z}{3z^2+x-2y}$

(D) $z+2xy-1$

(E) $-\frac{3z^2+x-2y}{x^2-2z}$

(F) $\frac{z+2xy-1}{3z^2+x-2y}$

(G) x^2-2z

(H) $-3z^2-x+2y$

(I) $-\frac{x^2-2z}{3z^2+x-2y}$

(J) $-\frac{z+2xy-1}{3z^2+x-2y}$

(K) $3z^2+x-2y$

(L) $-\frac{x^2}{3z^2+x}$

$$F(x,y,z) = z^3 + zx + x^2y - x - 2yz = 0$$

$$F_x = z + 2xy - 1$$

$$F_z = 3z^2 + x - 2y$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{z+2xy-1}{3z^2+x-2y}$$

6. (10 points) Find the critical point(s) of $f(x, y) = 13x^2 - 2x^3 + y^2 - 2xy$. Write your final answer(s) on the line provided.

Critical point(s): (0,0) , (4,4)

Show work here:

$$\begin{cases} (1)\ f_x = 26x - 6x^2 - 2y = 0 \\ (2)\ f_y = 2y - 2x = 0 \Rightarrow 2y = 2x \Rightarrow y = x \end{cases}$$

Substitute $y=x$ into (1): $26x - 6x^2 - 2x = 0$

$\Rightarrow 24x - 6x^2 = 0$

$\Rightarrow 6x(4-x) = 0$

$\Rightarrow x=0,\ x=4$

When $x=0$ from (2) $y=0$, so $(0,0)$ is a critical pt

when $x=4$ from (2) $y=4$, so $(4,4)$ is a critical pt

7. (10 points) Let $P = (-4, 1)$ be a critical point of a function $f(x, y)$, and let $Q = (1, 2)$ be a critical point of a function $g(x, y)$. Assume that $f_{xx}(P) = 2$, $f_{xy}(P) = 0$, $f_{yy}(P) = 4$ and $g_{xx}(Q) = 2$, $g_{xy}(Q) = -1$, $g_{yy}(Q) = 0$.

(7.1) Apply the second derivative test to classify P. Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

- (A) f has a local maximum at P.
- (B) f has a absolute maximum at P.
- (C) f has a local minimum at P.
- (D) f has a absolute minimum at P.
- (E) f has a saddle point at P.
- (F) Second derivative test is inconclusive at P.

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(P) = \begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix} \; 2(4) - 0^2 = 8 \underbrace{> 0}_{\text{must include}} \text{ and } f_{xx}(P) = 2 \underbrace{> 0}_{\text{must include}} \; \cup$$

or $f_{yy}(P) = 4 > 0$

or $D(P) = f_{xx}(P) f_{yy}(P) - (f_{xy}(P))^2$

(7.2) Apply the second derivative test to classify Q. Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

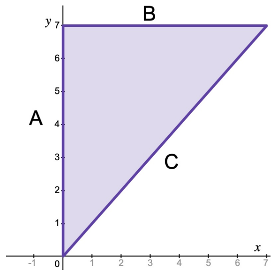
- (A) g has a local maximum at Q.
- (B) g has an absolute maximum at Q.
- (C) g has a local minimum at Q.
- (D) g has a absolute minimum at Q.
- (E) g has a saddle point at Q.
- (F) Second derivative test is inconclusive at Q.

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(Q) = \begin{vmatrix} 2 & -1 \\ -1 & 0 \end{vmatrix} = 2(0) - (-1)^2 = -1 \underbrace{< 0}_{\text{must include}}$$

8. (12 points) Let $f(x,y) = x^2 + y^2 - 2x - 12y$ on the domain D such that $x \geq 0$, $0 \leq y \leq 7$, and $y \geq x$ (see figure below).

(8.1) The maximum and minimum values of f on edge A have been provided. Find the maximum and minimum values of f on each of the remaining two edges of the triangle for the domain D. Complete the table with your final answers.



edge of triangle	A	B	C
max value of f	0	0	0
min value of f	−36	−36	$-\frac{49}{2}$

Show work below. You must provide justification for your answers for **each** remaining edge.

B: $y=7$, $0 \le x \le 7$: $f(x,7) = x^2+49-2x-84 = x^2-2x-35$

Let $g(x) = x^2-2x-35$

$g'(x) = 2x-2 = 0 \Rightarrow x = 1$

$\underbrace{g(0)}_{=f(0,7)} = -35$, $\underbrace{g(1)}_{=f(1,7)} = 1^2-2(1)-35 = -36$, $\underbrace{g(7)}_{=f(7,7)} = 7^2-2(7)-35 = 0$

C: $y=x$, $0 \le x \le 7$: $f(x,x) = x^2+x^2-2x-12x = 2x^2-14x$

Let $h(x) = 2x^2-14x$

$h'(x) = 4x-14 = 0 \Rightarrow x = 7/2$

$\underbrace{h(0)}_{=f(0,0)} = 0$, $h(7/2) = 2(7/2)^2 - 14(7/2) = \frac{49}{2} - \frac{98}{2} = -\frac{49}{2}$, $\underbrace{h(7)}_{=f(7,7) \text{ already found}} = 2(7)^2 - 14(7) = 0$

(8.2) The only critical point of the function f is $(1,6)$. Determine the absolute maximum and absolute minimum of value of f on the domain D. Put your answers in the spaces provided.

The absolute max value of f is ___0___ and the absolute min value of f is ___−37___.

Show any additional work here:

$f(1,6) = 1^2+6^2-2(1)-12(6)$

$= 1+36-2-72$

$= -37$

9. (10 points) Consider the problem of finding the points on the circle $x^2 + y^2 = 32$ at which the product xy is a maximum. Fill in the blanks below to set up this problem using the method of Lagrange multipliers.

(9.1) The function you need to maximize is $f(x, y) =$ xy

(9.2) The system you need to solve using Lagrange multipliers is a system with 3 (#) equations and 3 (#) unknowns.

(9.3) Using the method of Lagrange multipliers, write out the system of equations you need to solve for this problem. You **DO NOT** need to solve the system. Write your equations in the box below. (Do not just write the general equations.)

(1) $y = \lambda(2x)$

(2) $x = \lambda(2y)$

(3) $\underbrace{x^2+y^2-32=0}_{\text{or } x^2+y^2=32}$

OR

$\langle y, x\rangle = \lambda\langle 2x, 2y\rangle$

$x^2+y^2-32=0$

Note: $\nabla f = \lambda \nabla g \Rightarrow \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$ must follow from what was given for $f(x,y)$ in 9.1 and the constraint equation in 9.3.

10. (10 points) For each true/false question, fill in the appropriate bubble. No work required.

(10.1) ○ T ● F If a function $f(x, y)$ has continuous third-order partial derivatives then $f_{xyy} = f_{xxy}$.

(10.2) ○ T ● F If $\nabla f(a, b) = \langle 0, 0\rangle$, then f must have a local maximum or minimum at (a, b).

(10.3) ○ T ● F If $f(x, y) = x \ln(x + y)$, then $(0, 0)$ is a critical point of f. f not defined at (0,0)

(10.4) ● T ○ F The gradient $\nabla F(a, b, c)$ is perpendicular to the tangent plane to the level surface $F(x, y, z) = k$ at the point $P = (a, b, c)$.

(10.5) ● T ○ F For a function $f(x, y)$ and point $P = (a, b)$: $f_x(a, b) = D_{\mathbf{u}} f(a, b)$ when $\mathbf{u} = \langle 1, 0\rangle$.